Agent Orchestration

Model Selection Strategy

Day 1 - Session 1, Topic 2

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What We'll Cover

1. Models, agents, and selection tradeoffs
2. Task evidence, model tier, and effort
3. Completed-task economics
4. Demo: Model Selection Matrix
5. Demo: Evidence-Based Escalation
6. Interactive Model Selection Map

Model, Agent, and Orchestrator

A model is one component of an agent, not the entire agentic system.

REASONING ENGINE

Model

Produces reasoning and responses from supplied context.

GOAL-DIRECTED WORKER

Agent

Combines a model with instructions, tools, and an action loop.

COORDINATOR

Orchestrator

Assigns work, selects resources, coordinates dependencies, and checks outcomes.

Why Select Models?

Select for reliable completion by balancing four constraints, not by naming a permanent “best model.”

CAPABILITY

Ambiguity, breadth, and difficult reasoning.

SPEED

Latency and responsiveness.

COST

Usage price and resource consumption.

FIT

Performance with the required context and tools.

Route from Evidence

Capability, speed, and price are constraints, not labels

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Start with the Task Card

Score three signal groups before opening the model selector.

CAN IT SOLVE THIS?

Difficulty

Complexity, repository breadth, uncertainty, and missing context.

WHAT CAN GO WRONG?

Risk

Blast radius and reversibility.

WHAT IS COMPLETION WORTH?

Economics

Verification cost, available checks, latency, and completion value.

Model Selection Is Not Risk Control

A stronger model may reduce some execution errors.

It does not replace scoped permissions, security gates, or human approval.

High-blast-radius work needs stronger controls regardless of model choice.

> Capability answers “Can it reason about this?” Governance answers “May it act?”

Governance or Capability?

A strong model proposes a high-blast-radius change. Which factor is governance?

A) Context capacity

B) Approval authority

C) Debugging skill

D) Response speed

> **Chat Check:** Type your answer (A, B, C, or D) in the chat window now.

Governance or Capability: The Answer

B - Approval authority

Governance determines whether the model may act. Capability, context, and speed do not grant permission for a high-risk change.

Evidence-Based Model Routing (SaaS)

> **SaaS**: Software as a Service

REAL-WORLD SCENARIO

SaaS (Vercel - AI Gateway):

- Teams can route by measured uptime and latency, set provider order, and define model fallbacks.
- Model metadata exposes pricing and reasoning controls, while scoped budgets cap spend before each request.

Current Claude Model Baseline

Model	Typical starting point	Relative speed
Haiku 4.5	Routine, high-volume, cost-sensitive work	Fastest
Sonnet 5	Everyday coding with speed and capability	Fast
Opus 5	Complex agentic coding and enterprise work	Moderate
Fable 5.1	Demanding, long-horizon agentic work	Slower

> Verified 2026-09-13. Availability and product details can change.

Practical Model Selection (Claude-Helper)

REAL-WORLD SCENARIO

Start with the cheapest, fastest model that can complete the task. Escalate for writing or deeper reasoning.

READS AND SEARCH

Haiku

File reads, grep, find, and status checks; roughly 80% of tasks.

DEFAULT WRITER

Sonnet

Code changes, reviews, and debugging.

CRITICAL WORK

Opus

Architecture, security analysis, and failure recovery.

Effort Is a Separate Lever

Effort controls response-token use across text, thinking, and tool calls.

EVALUATED BOUNDARIES

Low or Medium

Use for bounded work after evaluations show quality holds.

DOCUMENTED DEFAULT

High

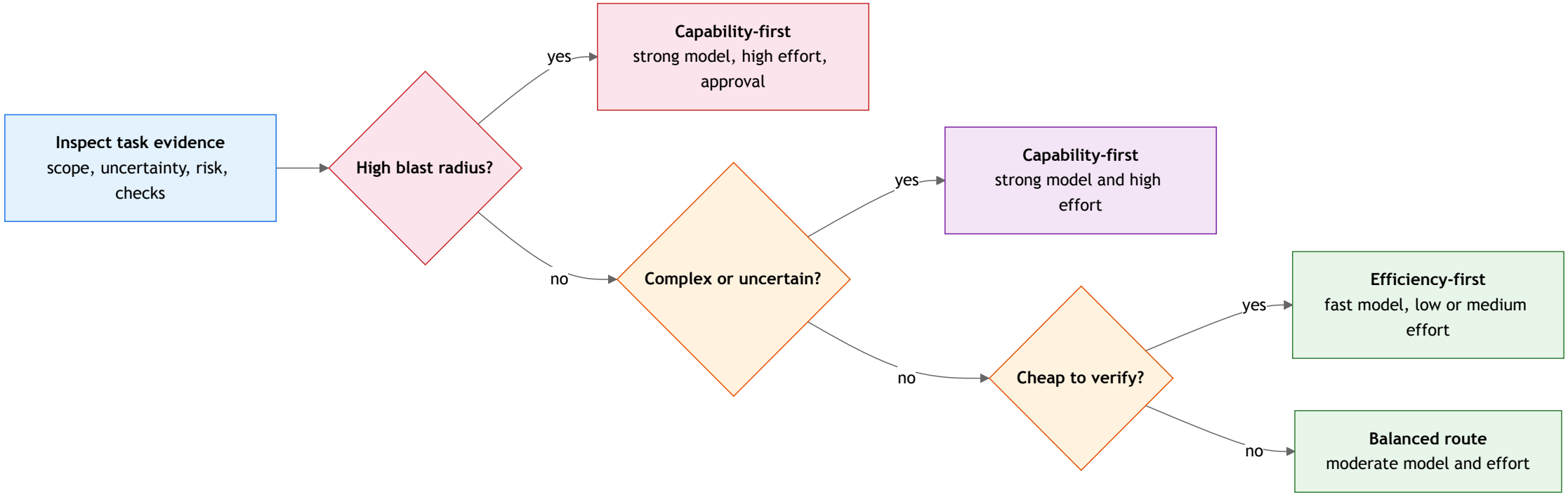
Use for difficult coding and agentic tasks.

EVIDENCE REQUIRED

Higher Levels

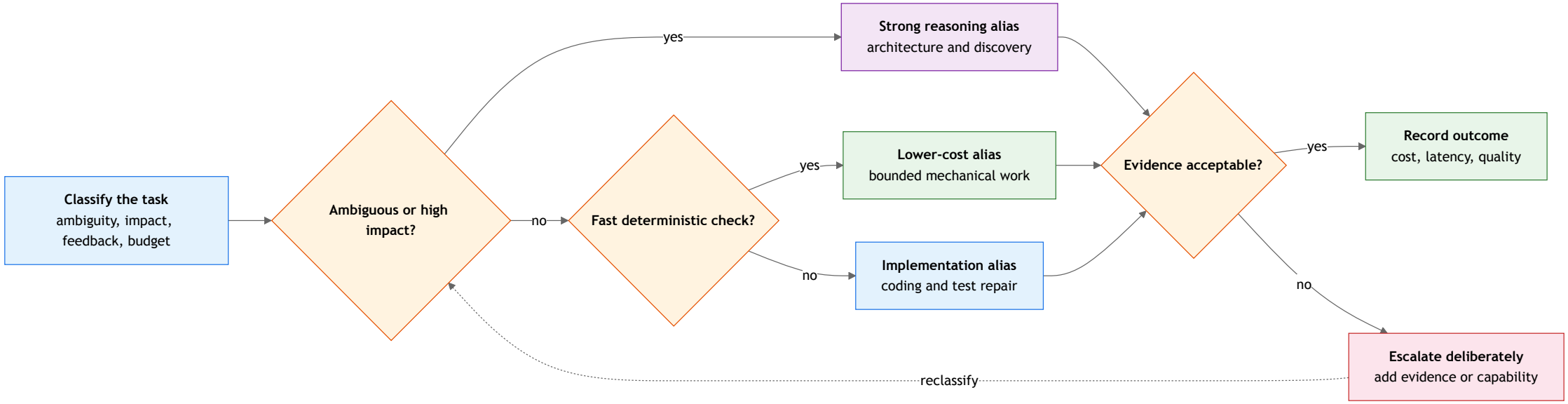
Use for supported, justified long-running work.

Model Selection Decision Tree



Generic Model Routing Decision

REFERENCE ONLY



Four Task Shapes

Task	Dominant signal	Initial strategy
Update delivery copy	Local and reversible	Efficiency first
Add a carrier mapping	Specified and easy to check	Balanced or efficient
Diagnose duplicates	Uncertain and cross-service	Capability first
Change refund authorization	High blast radius	Strong model plus approval

Demo: Model Selection Matrix

From the companion repo, run `python3 day1/concepts/demos-conceptual/demo-model-selection-matrix.py`.

- Evidence score: four task signals drive the initial route.
- Model and effort: both choices appear in the record.
- Approval: irreversible work remains human-controlled.

> Dated recommendations run offline; they do not call a model.

Inspect the Recommendation

The matrix is a governance aid, not a capability guarantee.

Ask whether each input came from repository evidence or assumption.

Check whether the recommended verification can detect a bad outcome.

Override the route when risk policy or organizational access requires it.

Optimize for Accepted Outcomes

Price the retries, review, and rework

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Per-Token Price Is Incomplete

Measure accepted work, not token price alone. Avoided failures can make a capable model cheaper per completion.

RELIABILITY

Attempts and pass rate.

TIME

End-to-end latency and human review.

USAGE

Input and output tokens.

REPAIR

Rework and regression repair.

Compare under Controlled Conditions

Keep the task, prompt, tools, sandbox, time, and resource limits fixed.

Run multiple trials when behavior varies.

Grade repository outcomes rather than the agent's completion message.

Anthropic measured a six-point benchmark swing from infrastructure alone.

Use the Right Grader

Evidence	Suitable grader
Tests, compilation, schema, or static checks	Code-based
Instruction following or nuanced quality	Calibrated rubric or model-based
Security, architecture, or business risk	Human review

Combine graders when no single check covers the acceptance contract.

Escalation Needs a Signal

TRIGGER

Reproducible failure, unexpected scope, missing context, or new risk.

CHANGE

One lever: effort, model, context, task boundary, or human control.

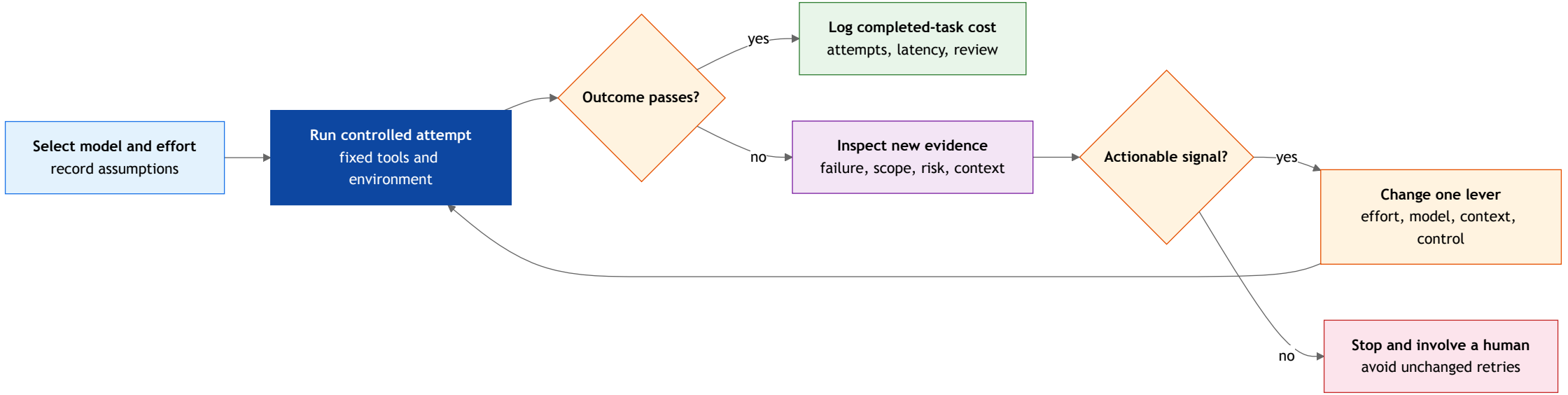
AVOID

Repeating unchanged conditions and calling the retry a strategy.

STOP

When the trajectory provides no actionable evidence.

Evidence-Based Escalation Loop



Predict the Routing Decision

The focused test passes, but a full scan shows that input sanitization was removed. What should happen next?

A) Stop for human security review
B) Upgrade the model automatically
C) Lower effort and retry
D) Repeat the unchanged prompt

> Decide privately first. Then compare reasons and produce one evidence-based answer.

Automated Escalation Loops

TRICK OF THE TRADE

In production, you don't manually check a test and re-prompt the agent. You write a script.

```
for attempt in {1..3}; do
  python3 -m unittest tests.test_security > result.log 2>&1
  if [ $? -eq 0 ]; then
    echo "Tests passed!"
    break
  fi
  echo "Tests failed. Feeding errors back to agent..."
  claude -p "Fix this test failure: $(cat result.log)"
done
```

Pro Tip: The orchestrator's job is to pipe `stderr` back to the agent automatically up to a maximum threshold.

Explain the Security Stop

A - Stop for human security review

The repository-wide evidence shows a crossed security boundary. A stronger model does not grant authority to continue automatically.

After the file is repaired and the same security check passes, the evidence supports `KEEP`.

> Debrief: Why would option B improve capability without resolving governance?

Demo: Interactive Model Selection

Open `day1/demos/interactive-model-selection.html` from the companion repository.

- Route tasks using complexity, risk, evidence, model, and effort signals.
- Inspect the mandatory human gate for high-risk work.
- Grade the outcome and escalate one lever when new evidence justifies it.

Select nodes and relationships to inspect their upstream and downstream context.

> Built with Archify from `interactive-model-selection.workflow.json`. Self-contained; no server, install, API key, or network required.

Demo: Evidence-Based Escalation

REFERENCE ONLY

From the companion repo, run `python3 day1/concepts/demos-conceptual/demo-evidence-based-escalation.py`.

- Focused failure: keeps the initial route while narrowing inspection.
- Cross-service scope: escalates capability deliberately.
- Security boundary: stops execution for human review.

> Reference alternative: use this deterministic trajectory instead of the practical prediction sequence, not before it.

Selection Record

ROUTE

Model, effort, date, entitlement, and fallback.

SIGNALS

Task evidence and controlled evaluation conditions.

RESULTS

Acceptance, attempts, latency, review, and total cost.

ESCALATION

Name the trigger before execution begins.

Key Takeaways

1. Select models from task evidence rather than task wording alone.
2. Tune effort separately and validate every reduction against evaluations.
3. Compare cost per accepted task, including retries and review.
4. Hold the harness constant when comparing configurations.
5. Escalate on new evidence and preserve human control for high risk.

Ready for Kahoot Quiz 1?

Choose the route, then defend it with evidence.

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